

Task 3: Adversity Prediction Model

Model Choice and Rationale

I used **horizon-specific classification models**, a separate model was trained for each horizon: 5, 10, 15, 20, 25, and 30 seconds.

Why? Adversity behavior can change with the closing horizon. For example, a trade that is not adverse after 5 seconds may become adverse after 30 seconds due to later price movement.

For each horizon, two models were trained and compared for each tau

- HistGradientBoostingClassifier
- ExtraTreesClassifier

Main choosing metric: The final model for each horizon was selected based on **validation log loss** since probability prediction was the main result needed, not just binary classification.

A lower log loss indicates better probability quality and calibration, which is important because Task 4 uses these predicted probabilities for externalization threshold decisions.

Mostly Tree based models were selected in this process since they provided much better results on this shirt dataset compared to the other model

Feature Selection and Justification

The model uses only information available at or **before the trade time**.

NOTE: Future mid-price columns such as M5, M10, M15, M20, M25, and M30 were used only for creating labels and were not used as model features.

The feature set includes **trade-level, time-based, market-state, and train-only client-history features**.

Trade-level features include **side, volume, trade price, M0, spread, price versus mid, relative spread, signed volume, and log volume**. (Showcase how aggressively the trade was executed relative to the mid price)

Time-based features include **hour, minute, second, day of week, and seconds from midnight**. (intraday trading patterns and possible changes in flow behavior over the trading day)

Market-state features include recent M0 changes, past volatility, past signed flow, and past spread averages. These are computed within each trading date. Rolling volatility, rolling signed flow, and rolling spread features use only previous trades through a one-step shift, so the current trade is not included in its own history feature. This avoids look-ahead bias.

Client-history features include train-only adverse rates by client, client and horizon, client and side, client-side-horizon, and side-horizon. These features capture historical adverse-flow tendencies but are computed only from the training set. The same train-derived tables are then applied to validation and test sets. This avoids leakage from validation or test labels.

Results

The final metrics are reported in `task3\_results.csv` with rows for train, validation, and test. The required metrics are accuracy, precision, recall, and log loss

Final results: